

Forecasting Respiratory Virus Activity with Cross-Pathogen Information

A Multi-Model Regional Forecasting Project Report

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This report consolidates the finalized forecasting outputs for influenza, respiratory syncytial virus (RSV), and SARS-CoV-2 across the primary regional forecasting settings, with an additional Singapore case study. The supplied final figures are embedded directly in the manuscript and are treated as the authoritative result artifacts for the numerical and directional findings reported here.

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Executive Summary

This project evaluates 1-, 2-, and 4-week forecasts for influenza, RSV, and SARS-CoV-2, emphasizing whether other-pathogen activity adds useful predictive information. The primary regional outputs cover Brunei Darussalam, Indonesia, and Malaysia, with a dedicated Singapore influenza–SARS-CoV-2 case study.

The principal findings are:

- **Simple temporal baselines were difficult to beat consistently.** In the overall mean-MAE ranking, M2 Persistence ranked first (1.81), followed by M3 Persistence (2.11), M2 Recent Mean (2.33), and M3 Recent Mean (2.48). Among the learned models, XGBoost was strongest, with mean ranks of 3.19 (M3) and 3.22 (M2).
- **Cross-pathogen information was conditionally useful rather than universally beneficial.** The M3-versus-M2 comparison was positive for RSV and SARS-CoV-2 across the displayed model families, but mixed for influenza. Benefits were also more apparent at 2- and 4-week horizons than at 1 week.
- **Own-pathogen history remained the dominant predictive signal.** Grouped SHAP analysis assigned a median relative importance of 68.0% to own-pathogen history, compared with 16.0% to other-pathogen activity and 10.5% to seasonality.
- **The Singapore case study showed strong heterogeneity by pathogen and model.** The Negative Binomial model exhibited a very large positive M3-versus-M2 improvement for influenza across all displayed horizons, while its SARS-CoV-2 effects were slightly negative. XGBoost effects were modestly positive in several short-horizon settings but strongly negative for 4-week SARS-CoV-2 forecasting.
- **Probabilistic calibration was sensitive to the interval construction strategy.** Primary intervals achieved observed test-set coverage above the nominal 80% line, while the validation-calibrated alternative substantially under-covered in both M2 and M3, indicating that calibration choices require careful out-of-sample checking.

Overall, cross-pathogen signals add value selectively, especially at longer short-term horizons, but should augment rather than replace strong own-pathogen history and robust temporal baselines.

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Chapter 1

Introduction

Respiratory-virus surveillance is intrinsically multivariate. Influenza, respiratory syncytial virus (RSV), and SARS-CoV-2 can circulate in the same population, exhibit distinct seasonal patterns, and vary in their temporal relationship across countries and periods. This creates a forecasting question that is both epidemiologically and operationally relevant: *does recent activity in other pathogens improve short-horizon forecasts beyond a target pathogen’s own recent history and seasonal structure?*

This project addresses that question using a layered forecasting workflow. The finalized analysis compares simple temporal baselines with a Negative Binomial model, gradient-boosted trees (XGBoost), and an ensemble; evaluates models with mean absolute error (MAE); uses SHAP-based grouped attribution to interpret XGBoost forecasts; examines cross-pathogen effects by target pathogen and forecast horizon; assesses probabilistic interval calibration; and presents a focused Singapore influenza–SARS-CoV-2 case study. XGBoost and SHAP are well-established tools for nonlinear prediction and post-hoc feature attribution, respectively [1, 2].

The report has four main objectives:

1. describe the seasonal and forecasting structure represented in the final project outputs;
2. compare model families and establish an overall forecasting performance hierarchy;
3. determine when cross-pathogen information improves or worsens MAE; and
4. interpret the relative contribution of own-pathogen history, other-pathogen activity, and seasonality while checking uncertainty calibration.

A central theme throughout the report is that forecasting gains are heterogeneous. A model or feature group that helps at one horizon or for one pathogen may fail to help elsewhere. The project therefore emphasizes comparative evidence rather than a single universally best specification.

Chapter 2

Study Design and Forecasting Setting

2.1 Pathogens and geographic settings

The finalized outputs analyze three respiratory-virus targets: influenza, RSV, and SARS-CoV-2. The primary regional seasonality figure contains Brunei Darussalam, Indonesia, and Malaysia. A separate Singapore case study focuses on influenza and SARS-CoV-2. The response plotted in the seasonal analysis is weekly *mean relative activity*, indexed by ISO epidemiological week.

2.2 Forecast horizons

The cross-pathogen horizon analysis and Singapore case study explicitly evaluate 1-week, 2-week, and 4-week forecasting horizons. This design permits examination of a common forecasting trade-off: very short-horizon predictions may be dominated by persistence, whereas longer horizons can potentially benefit more from broader contextual predictors.

2.3 M2 and M3 comparison

The final result figures use the labels M2 and M3 and report “M3 vs M2 MAE improvement (%)”. The cross-pathogen figures establish this contrast as the project’s test of added cross-pathogen information. Accordingly, M2 is treated as the reference forecasting specification and M3 as the cross-pathogen-augmented specification. Positive values indicate lower MAE for M3 than M2; negative values indicate worse MAE after adding the cross-pathogen information represented by M3.

Because the uploaded final figures do not encode the full feature-engineering recipe or every preprocessing setting, this report does not invent a more detailed M2/M3 formula. The interpretation is kept at the level directly supported by the finalized outputs.

2.4 Seasonal heterogeneity

Figure 2.1 illustrates substantial heterogeneity in weekly activity profiles across pathogens and countries. Some series show relatively smooth within-year changes, while others contain sharper peaks, including pronounced end-of-year increases in selected country–pathogen series. The visual differences motivate the use of both own-pathogen lagged information and explicit seasonal features rather than assuming one common seasonal pattern for all settings.



Figure 2.1: Seasonal respiratory-virus activity in the primary regional forecasting settings. The figure shows weekly mean relative activity for influenza, RSV, and SARS-CoV-2 where available in Brunei Darussalam, Indonesia, and Malaysia.

Chapter 3

Forecasting and Evaluation Methods

3.1 Baseline forecasts

Three simple temporal baselines appear in the final model comparison:

- **Persistence**: carries the latest observed level forward.
- **Recent mean (4 weeks)**: uses a short rolling average to reduce sensitivity to one unusually high or low week.
- **Seasonal naive (52 weeks)**: uses the corresponding value from approximately one year earlier.

These models provide essential reference points. In short-horizon infectious-disease forecasting, a complex model is only operationally compelling if it improves on strong low-variance temporal rules.

3.2 Negative Binomial model

The project includes a Negative Binomial forecasting model. Negative Binomial regression is commonly used for overdispersed surveillance outcomes because it relaxes the equal-mean-and-variance restriction of a Poisson model [3]. In this report, the model is interpreted as the project's parametric count-oriented forecasting component. The exact link function, regularization choices, and feature matrix are not recoverable from the supplied final figures and therefore are not restated beyond the finalized model label.

3.3 XGBoost

XGBoost provides a nonlinear tree-based model capable of representing interactions and non-additive effects among lagged activity, other-pathogen predictors, and seasonal variables [1]. Its

flexibility makes it a natural complement to the more structured Negative Binomial specification.

3.4 Ensemble

An ensemble model is included in the final comparisons. The supplied figure set establishes its performance but does not encode the precise weighting or combination rule. The report therefore treats the ensemble as a finalized forecast product and does not assume a weighting formula that is not present in the artifacts.

3.5 Performance metric and ranking

Forecast accuracy is evaluated using mean absolute error (MAE). MAE is

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (3.1)$$

where y_i is the observed outcome and $\hat{y}_i$ is the corresponding point forecast. Lower MAE indicates better accuracy.

The final overall comparison summarizes performance using *mean MAE rank*; lower ranks are better. Ranking is useful when aggregating performance across multiple forecasting settings because it reduces the dominance of any single high-scale series.

3.6 Cross-pathogen improvement

The primary effect measure for the M3-versus-M2 comparison is the percentage MAE improvement. The sign convention in the finalized figures is straightforward: values above zero favor M3 (cross-pathogen information improves accuracy), whereas values below zero favor M2.

3.7 SHAP interpretability

The project uses SHAP values to summarize XGBoost predictor contributions [2]. Features are aggregated into three conceptually meaningful groups:

1. own-pathogen history;
2. other-pathogen activity; and
3. seasonality.

The final figure reports median relative SHAP importance, allowing the project to distinguish predictive dominance from mere feature availability.

3.8 Probabilistic forecasts and calibration

The supplementary analysis compares observed test-set coverage of forecast intervals against a nominal 80% target. Two interval strategies are displayed: the project's primary intervals and a validation-calibrated sensitivity alternative. A well-calibrated 80% interval should cover approximately 80% of held-out observations over repeated forecasts; systematic over-coverage implies overly wide intervals, while under-coverage implies intervals that are too narrow.

Chapter 4

Results

4.1 Overall model ranking

Figure 4.1 shows the aggregate forecasting hierarchy. The two best mean MAE ranks belong to Persistence (M2: 1.81; M3: 2.11), followed by the 4-week Recent Mean (M2: 2.33; M3: 2.48). This is an important result: simple, stable temporal baselines remained highly competitive across the aggregate evaluation.

Among the learned models, XGBoost ranked best (M3: 3.19; M2: 3.22), followed by the ensemble (M3: 3.44; M2: 3.59). The Negative Binomial model ranked lower (M3: 4.30; M2: 4.41), while the 52-week seasonal-naive baseline ranked last (M3: 4.81; M2: 4.96).

Table 4.1: Overall forecasting model ranking reproduced from the finalized figure. Lower mean MAE rank is better.

Model variant	Mean MAE rank
M2 Persistence	1.81
M3 Persistence	2.11
M2 Recent mean (4 weeks)	2.33
M3 Recent mean (4 weeks)	2.48
M3 XGBoost	3.19
M2 XGBoost	3.22
M3 Ensemble	3.44
M2 Ensemble	3.59
M3 Negative Binomial	4.30
M2 Negative Binomial	4.41
M3 Seasonal naive (52 weeks)	4.81
M2 Seasonal naive (52 weeks)	4.96

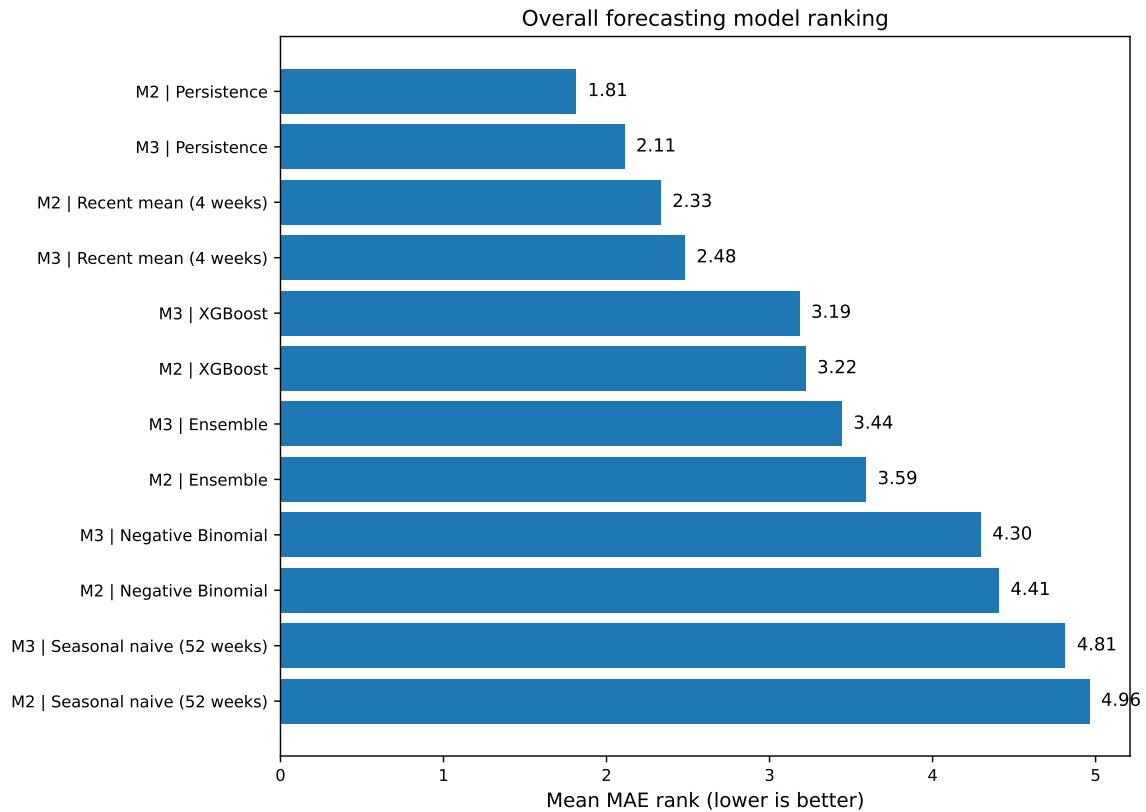


Figure 4.1: Overall forecasting model ranking based on mean MAE rank. Lower values indicate stronger aggregate performance.

4.2 Cross-pathogen effects by target pathogen

Figure 4.2 demonstrates that the value of cross-pathogen information depends strongly on the target pathogen. For RSV, all three displayed model families show positive median M3-versus-M2 MAE improvement. SARS-CoV-2 also shows positive improvement across the displayed models, with the largest gain for the ensemble and only a very small positive effect for XGBoost.

Influenza is qualitatively different. The ensemble and Negative Binomial bars are negative, with the Negative Binomial deterioration visually much larger, while XGBoost shows a small positive improvement. Cross-pathogen information therefore cannot be treated as uniformly beneficial across pathogens.

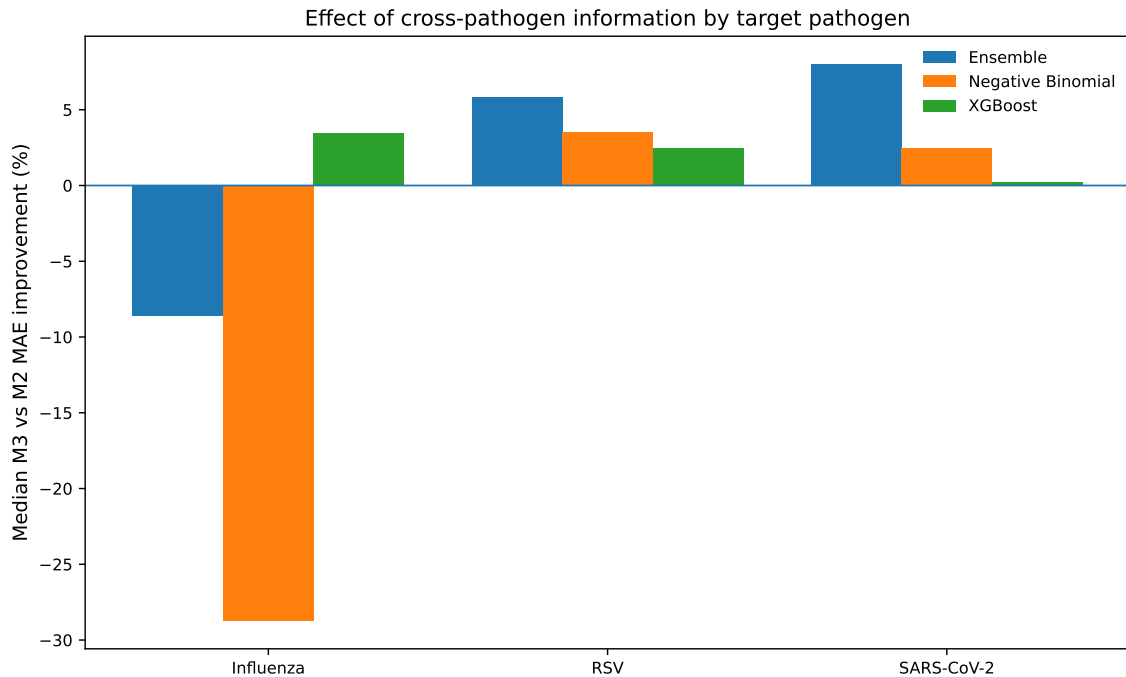


Figure 4.2: Median percentage MAE improvement of M3 over M2 by target pathogen. Positive values favor the cross-pathogen-augmented M3 specification; negative values indicate deterioration relative to M2.

4.3 Cross-pathogen effects by forecast horizon

The horizon analysis in Figure 4.3 reveals a clear temporal pattern. At the 1-week horizon, cross-pathogen information is mixed: the ensemble and XGBoost show negative median improvement, while the Negative Binomial model is only slightly positive. At 2 weeks, all three model families become positive. At 4 weeks, all three are again positive and the improvements are visually larger than at 2 weeks.

This result is consistent with a forecasting interpretation in which the target series' own recent value dominates the immediate next week, while broader system-level signals become more informative as the forecast horizon extends.

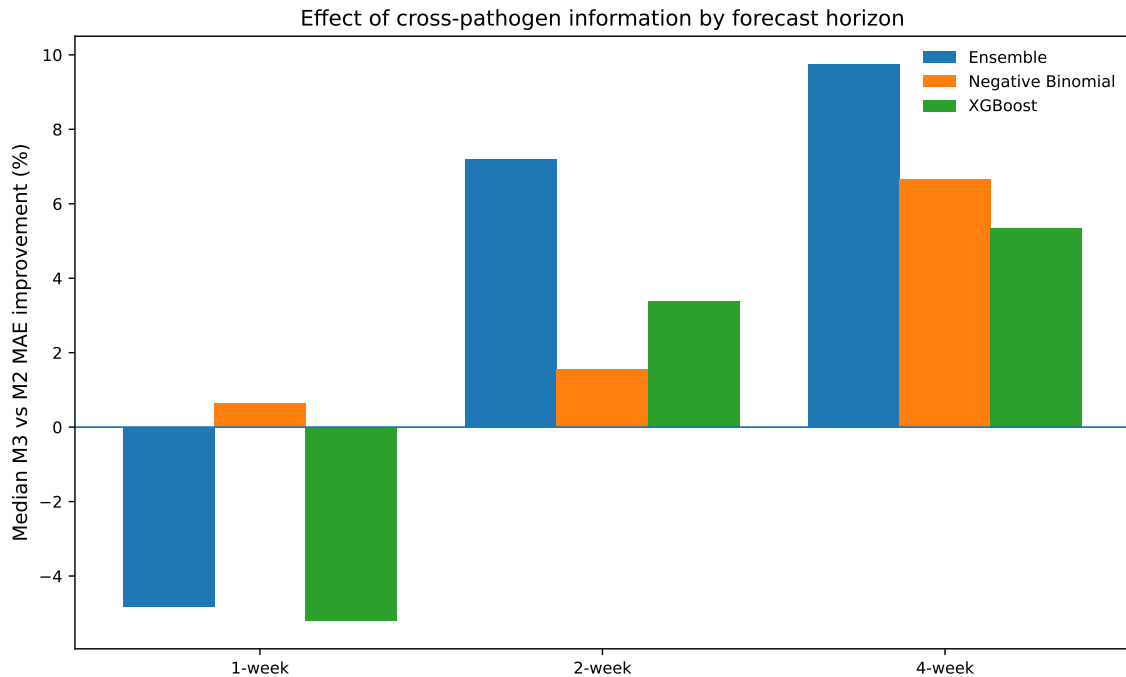


Figure 4.3: Median percentage MAE improvement of M3 over M2 by forecast horizon. Cross-pathogen information is mixed at 1 week but beneficial across the displayed model families at 2 and 4 weeks.

4.4 Grouped SHAP importance

The grouped XGBoost attribution results are shown in Figure 4.4. Own-pathogen history dominates with a median relative SHAP importance of 68.0%. Other-pathogen activity contributes 16.0%, while seasonality contributes 10.5%.

Table 4.2: Median relative SHAP importance of XGBoost predictor groups.

Predictor group	Median relative SHAP importance
Own-pathogen history	68.0%
Other-pathogen activity	16.0%
Seasonality	10.5%

The interpretation is not that other pathogens are unimportant; rather, cross-pathogen predictors provide a secondary signal that is meaningful but substantially smaller than the target pathogen’s own temporal history. This is compatible with the accuracy results, which show gains only in selected pathogens and horizons.

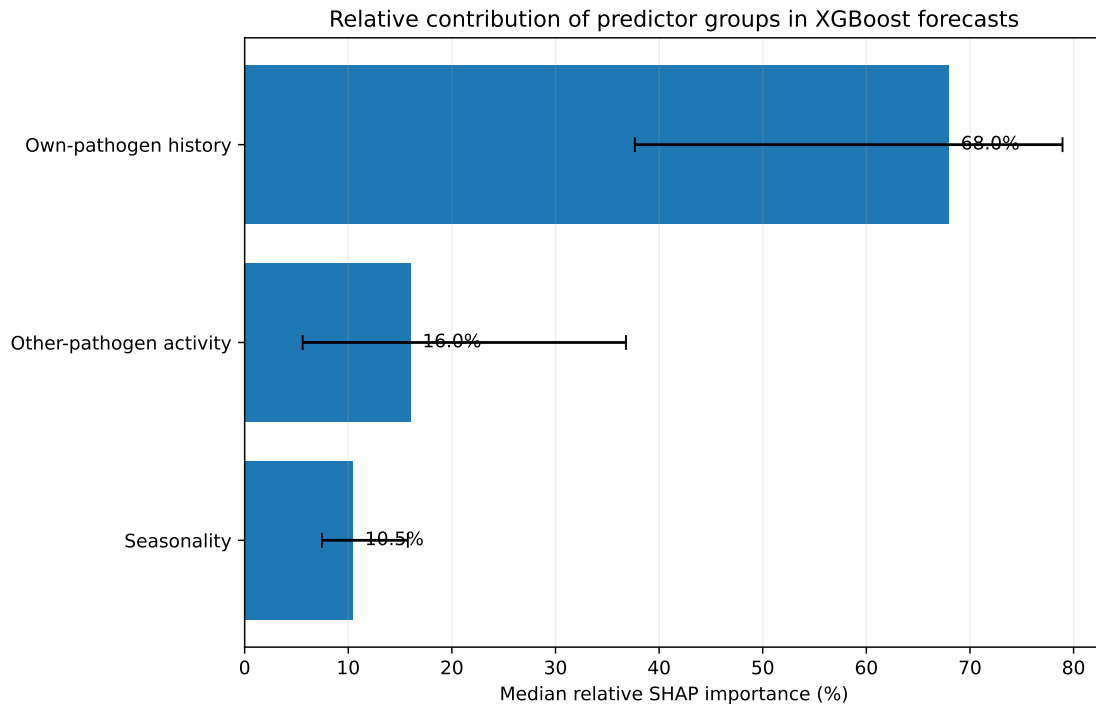


Figure 4.4: Relative contribution of predictor groups in XGBoost forecasts. Bars show median relative SHAP importance and error bars summarize variation across the underlying forecasting settings.

4.5 Singapore influenza–SARS-CoV-2 case study

Figure 4.5 provides a focused test of whether the aggregate cross-pathogen conclusions generalize to a specific setting. For influenza, the Negative Binomial model shows a strikingly large positive M3-versus-M2 improvement at all three horizons, visually around the mid-90% range. XGBoost shows smaller positive gains at 1 and 2 weeks and a slight negative effect at 4 weeks.

For SARS-CoV-2, the pattern reverses. The Negative Binomial M3-versus-M2 effect is slightly negative at each horizon. XGBoost is positive at 1 and 2 weeks but becomes strongly negative at 4 weeks. The case study therefore reinforces the main conclusion that cross-pathogen information is context dependent: it can be highly valuable in one target/model combination and harmful in another.

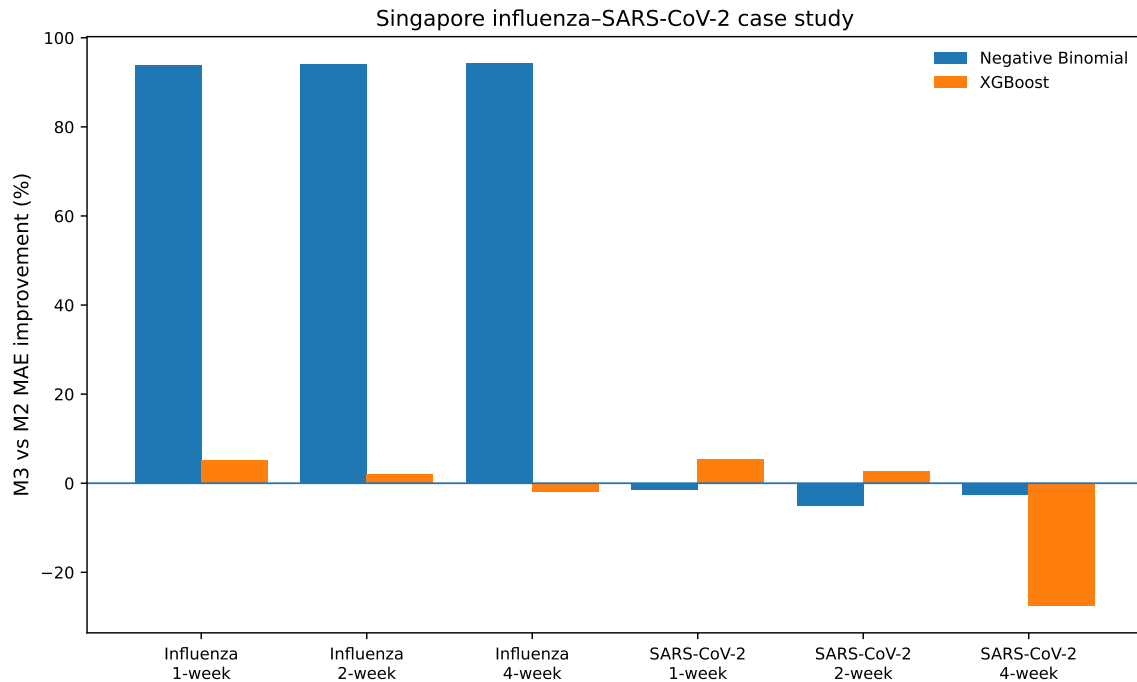


Figure 4.5: Singapore influenza–SARS-CoV-2 case study showing M3-versus-M2 MAE improvement for Negative Binomial and XGBoost forecasts at 1-, 2-, and 4-week horizons.

4.6 Probabilistic calibration sensitivity

The supplementary calibration result is shown in Figure 4.6. The primary intervals achieve observed test-set coverage above the nominal 80% target for both M2 and M3 (visually above 0.9), indicating conservative or relatively wide intervals. By contrast, the validation-calibrated sensitivity intervals cover only about 0.58–0.61 of test observations, substantially below the nominal target.

The sensitivity result shows that interval calibration is not automatically improved by an additional validation adjustment. A calibration rule that appears reasonable on a validation segment can transfer poorly to the test period, especially when the surveillance process is nonstationary.

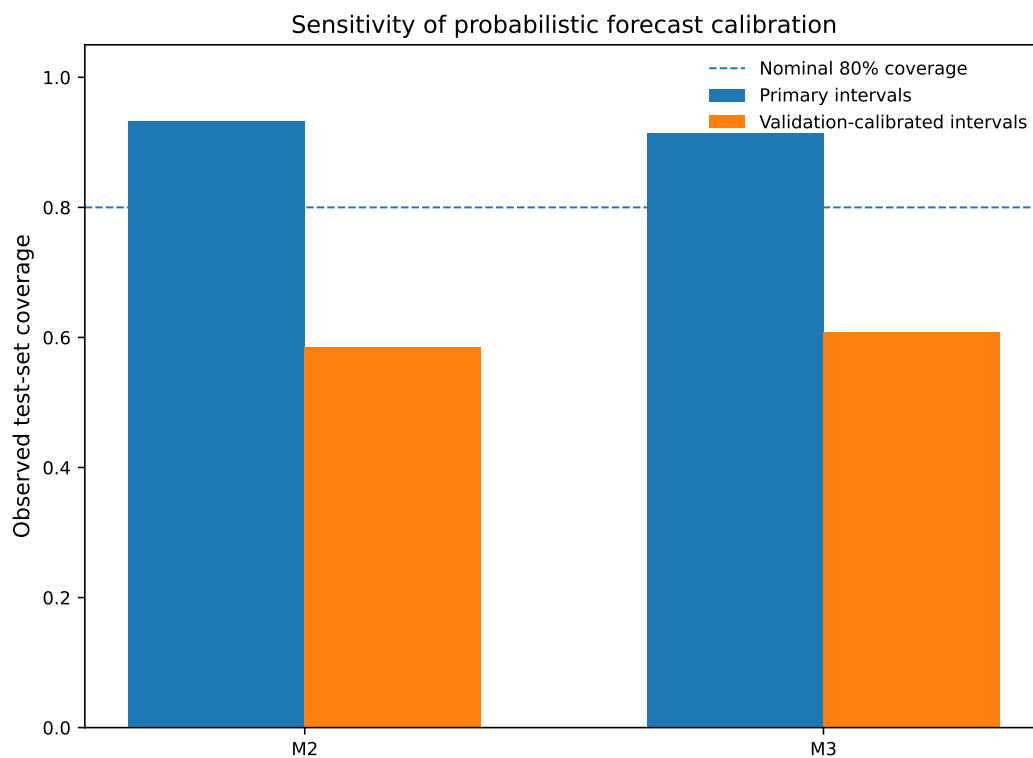


Figure 4.6: Sensitivity of probabilistic forecast calibration. The dashed line marks nominal 80% coverage; primary intervals over-cover, while the validation-calibrated alternative under-covers in both M2 and M3.

Chapter 5

Discussion

5.1 What the model ranking implies

The strongest aggregate result is the competitiveness of simple persistence and recent-mean forecasts. This should not be interpreted as a failure of machine learning. Instead, it indicates that weekly respiratory-virus activity contains strong local temporal continuity, and that a low-variance rule can be difficult to outperform when forecasts are only a few weeks ahead. The learned models still provide value by revealing when additional information matters and by enabling nonlinear cross-pathogen effects that simple baselines cannot represent.

XGBoost was the strongest learned model in the overall ranking, narrowly ahead of the ensemble and clearly ahead of the Negative Binomial specification. Its advantage is consistent with the ability of boosted trees to capture nonlinear thresholds and interactions without requiring a single prespecified functional form [1].

5.2 Cross-pathogen information is a conditional signal

The project does not support a blanket claim that adding other-pathogen activity always improves prediction. Instead, three interacting dimensions determine its value: target pathogen, forecast horizon, and model family.

First, RSV and SARS-CoV-2 show positive median improvement across the three model families in the pathogen-level summary, whereas influenza is mixed. Second, the horizon analysis shows the most coherent improvements at 2 and 4 weeks, while 1-week performance is mixed. Third, the Singapore case study shows that the same M3 augmentation can be extremely helpful for one model/pathogen combination and harmful for another.

This conditionality is scientifically plausible. Co-circulating pathogens may share seasonal drivers, behavioral changes, health-care seeking patterns, or surveillance artifacts, but their relationships are neither fixed nor necessarily causal. Cross-pathogen predictors should therefore be treated as contextual signals whose value must be validated out of sample.

5.3 Interpretability supports the forecasting results

Grouped SHAP attribution provides an important consistency check. Own-pathogen history accounts for 68.0% of median relative importance, much larger than the 16.0% contribution from other-pathogen activity. The attribution therefore agrees with the model-ranking result: immediate history is the primary source of forecast skill. At the same time, a 16% median contribution is large enough to matter, particularly when the forecast horizon is long enough for persistence to weaken.

The seasonality group contributes 10.5% of median relative SHAP importance. Figure 2.1 shows why this signal may be useful but insufficient on its own: seasonal profiles are visibly heterogeneous across countries and pathogens, and a fixed 52-week seasonal-naive forecast performs poorly in the overall ranking.

5.4 Why the Singapore case study matters

The Singapore analysis is valuable because aggregate summaries can hide setting-specific behavior. The very large influenza improvement for the Negative Binomial model suggests that, in this specific paired-pathogen setting, cross-pathogen variables can materially alter a structured model's forecast. However, the simultaneous deterioration for SARS-CoV-2 and the strongly negative 4-week XGBoost effect caution against transferring that result indiscriminately.

Operationally, this argues for pathogen-specific model selection and monitoring. A production pipeline could retain a strong baseline as a fallback, enable cross-pathogen features only where rolling validation supports them, and monitor whether the sign and magnitude of the benefit remain stable over time.

5.5 Uncertainty calibration requires separate validation

Point-forecast accuracy and interval calibration are distinct objectives. The supplementary result shows that a seemingly more deliberate validation-calibrated interval strategy can under-cover badly on the test set, while the primary intervals are conservative. This is a reminder that calibration procedures themselves can overfit. For surveillance forecasting, interval performance should therefore be reported alongside point metrics and re-evaluated as the data-generating process changes.

Chapter 6

Limitations

Several limitations should be considered when interpreting the project.

1. **Final-output provenance.** This report was assembled from the finalized figure artifacts supplied for the project. The uploaded set does not include the raw surveillance tables, preprocessing code, train/validation/test split definitions, hyperparameter grids, or the ensemble weighting rule. The report therefore avoids inventing those details and limits numerical claims to information visible in the final outputs.
2. **Association is not causation.** Improvement after adding other-pathogen activity does not establish biological interaction or causal interference between pathogens. Shared seasonality, common behavioral drivers, reporting patterns, or coincident outbreaks can create predictive association.
3. **Heterogeneity across settings.** The sign and magnitude of the M3-versus-M2 effect vary by pathogen, horizon, model, and geographic context. Aggregate performance can therefore obscure local failures.
4. **Baseline dominance.** Persistence and recent-mean baselines rank extremely well. Any practical deployment should compare advanced models against these baselines continuously rather than assuming a learned model will remain superior.
5. **SHAP interpretation.** SHAP describes how the fitted XGBoost model uses its predictors; it does not identify causal effects. Correlated lagged variables can also redistribute attribution across related features or groups [2].
6. **Calibration instability.** The large difference between primary and validation-calibrated test coverage indicates that interval calibration is sensitive to the chosen procedure and potentially to temporal nonstationarity.

These limitations do not negate the forecasting findings. They instead define the appropriate claim: cross-pathogen information is a potentially useful predictive signal that should be selected and validated conditionally, not assumed to provide universal improvement.

Chapter 7

Conclusion

This project developed and compared a multi-model framework for short-horizon respiratory-virus forecasting and used the final outputs to test the added value of cross-pathogen information. Five conclusions emerge.

First, simple temporal baselines remain formidable: persistence and a 4-week recent mean occupy the top positions in the overall mean-MAE ranking. Second, XGBoost is the strongest learned model in the aggregate comparison, followed by the ensemble and Negative Binomial model. Third, cross-pathogen information helps selectively rather than universally: gains are clearest for RSV and SARS-CoV-2 in the pathogen-level summary and at the 2- and 4-week horizons, while influenza and some 1-week settings show deterioration. Fourth, grouped SHAP analysis shows that the target pathogen’s own history remains the dominant predictive source, while other-pathogen activity supplies a meaningful secondary contribution. Fifth, the uncertainty sensitivity analysis demonstrates that interval calibration can fail out of sample even when point forecasting remains useful.

The practical message is therefore not to choose between autoregressive and cross-pathogen forecasting. A stronger strategy is to preserve robust temporal baselines, add cross-pathogen information where rolling validation supports it, and monitor both point accuracy and interval calibration separately. The Singapore case study illustrates exactly why such conditional deployment is necessary: the same cross-pathogen augmentation can deliver very large gains for one pathogen/model pair and meaningful deterioration for another.

Bibliography

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